



Bansilal Ramnath Agarwal Charitable Trust's
Vishwakarma Institute of Information Technology

**Department of
Artificial Intelligence and Data Science**

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Subject Name & Code: Cloud and DevOps

Assignment No: 03

Title of Assignment: Execute basic Linux commands (file operations, process management, user management)

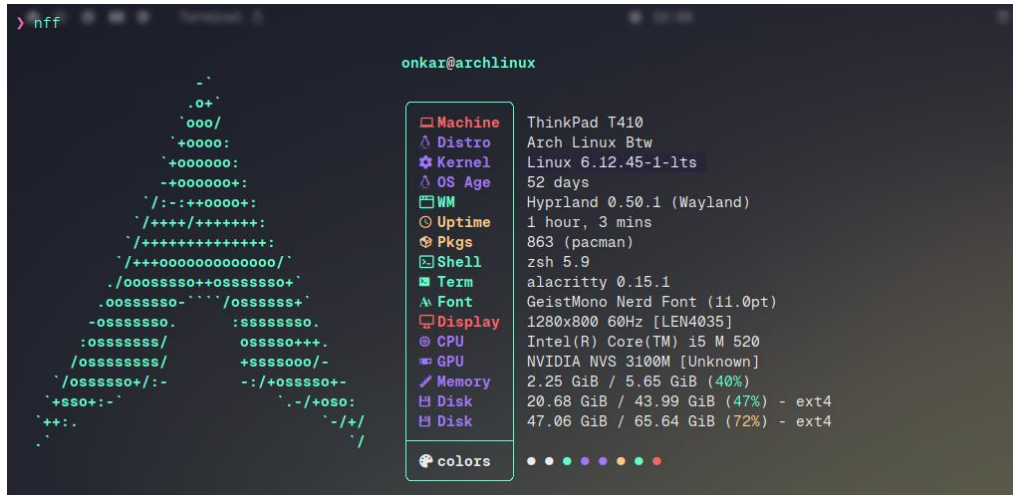
Date of Performance: / /2025

Date of Submission: / /2025

Aim: To execute Linux commands (file operations, process management, user management)

Objective: To learn basic Linux commands like file operations, process management and user management

Materials Required: Laptop with Arch Linux(OS kernel Linux 6.12.45-1-lts) installed in it, zsh(highly configured .zshrc).



The image shows a terminal window with a dark background. On the left, there is a large ASCII art logo made of characters like '.', '+', and '/'. On the right, there is a system information panel with a green border. The panel lists various system details:

Category	Value
Machine	ThinkPad T410
Distro	Arch Linux Btw
Kernel	Linux 6.12.45-1-lts
OS Age	52 days
WM	Hyprland 0.50.1 (Wayland)
Uptime	1 hour, 3 mins
Pkgs	863 (pacman)
Shell	zsh 5.9
Term	alacritty 0.15.1
Font	GeistMono Nerd Font (11.0pt)
Display	1280x800 60Hz [LEN4035]
CPU	Intel(R) Core(TM) i5 M 520
GPU	NVIDIA NVS 3100M [Unknown]
Memory	2.25 GiB / 5.65 GiB (40%)
Disk	20.68 GiB / 43.99 GiB (47%) - ext4
Disk	47.06 GiB / 65.64 GiB (72%) - ext4
Colors	• • • • • • • • • •

Procedure:

File Operations: File Operations are tasks performed on files and directories such as creating, copying, moving, and deleting.

□ Commands used:

- o touch – Create a file
- o cp – Copy files/directories
- o mv – Move or rename
- o rm – Remove
- o ls – List contents
- o cat, more, less – View file contents
- o chmod, chown – Manage permissions and ownership

Purpose:

To organize data, manage storage, and control access.

Step 1: Open terminal

Step 2: Create a folder

Step 3: Create a new text file

Step 4: Rename the text file

Step 5: Write text in the file

Step 6: Delete the file and folder

```

Terminal
12-Sep 180% 40%
> ps
  PID TTY          TIME CMD
 14289 pts/0    00:00:01 zsh
 143522 pts/0    00:00:00 zsh
 143527 pts/0    00:00:00 gitstatusd-linu
 145921 pts/0    00:00:00 ps

> who
onkar    tty2            2025-09-12 21:05
onkar    pts/1          2025-09-12 21:13 (tmux(19195).%0)
onkar    pts/0          2025-09-12 22:10 (tmux(19195).%1)
onkar    pts/3          2025-09-12 22:10 (tmux(19195).%2)
onkar    pts/4          2025-09-12 22:10 (tmux(19195).%3)

~ >
10:10:34 PM

> mkdir Temp$$
> ls
DATABASE      Lienzo-Backend      recovery
db             'Lienzo-Backend (2)' SIH25
DB             'Lienzo-Backend (3)' Temp145250
Desktop       'Lienzo-Backend (4)' test
Documents     'Lienzo-Backend (5)' Videos
dotfiles      mongo_data
Downloads     Music
DSA           node_modules
FastShop      Pictures
frontend-better-auth Practical
git-clones   Practice
inbox        Projects
Learn        react-playground

~ >
10:10:51 PM

cat server.js
File: server.js
1  require("dotenv").config();
2  const express = require("express");
3  const ConnectMongo = require("../config/conn/db.js");
4  const app = express();
5
6  // Connect to database
7  ConnectMongo();
8
9  // Setup all middleware
10 const setupMiddleware = require("../api/middleware/index.
    middleware.js");
11 setupMiddleware(app);
12
13 // Routes
14 const products = require("../api/routes/product.routes.js
    ");
15 const home = require("../api/routes/home.routes.js");
16
17 app.use("/products", products);
18 app.use("/", home);
19
20 app.listen(process.env.PORT, () => {
21   console.log( Listening on port: ${process.env.PORT}
22 );
23 });

~/FastShop/Backend main* >
10:11:18 PM
UP: 1h 5m
```

- o adduser / useradd – Add a new user
- o passwd – Set or change user password
- o deluser / userdel – Delete a user
- o usermod – Modify a user account

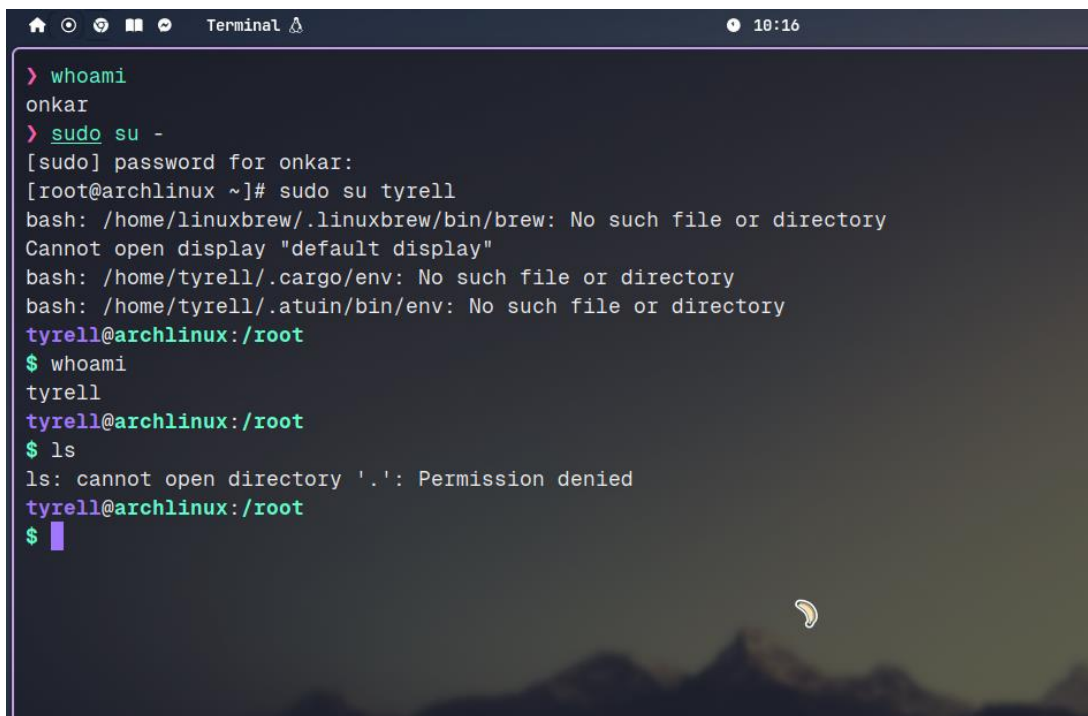
□ Groups are used to manage permissions for multiple users together. Purpose:
To provide personalized environments and restrict access based on roles.

Step 1: Open Terminal

Step 2: Command to display username

Step 3: Command to add a user

Step 4: Command to add new
password of the user



```

> whoami
onkar
> sudo su -
[sudo] password for onkar:
[root@archlinux ~]# sudo su tyrell
bash: /home/linuxbrew/.linuxbrew/bin/brew: No such file or directory
Cannot open display "default display"
bash: /home/tyrell/.cargo/env: No such file or directory
bash: /home/tyrell/.atuin/bin/env: No such file or directory
tyrell@archlinux:/root
$ whoami
tyrell
tyrell@archlinux:/root
$ ls
ls: cannot open directory '.': Permission denied
tyrell@archlinux:/root
$
  
```

Conclusion:

Though this assignment I explored the Linux OS(Arch BTW), where I explored to manage everything from cursor to window manager and config files of each individual application I am using(Neovim, zsh, tmux and etc) Along with that I learned to manage users, basic commands and etc.